

LegalEase

AI-Powered Legal Document Generator

Project Documentation

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1. Introduction

LegalEase is a web application that uses Generative AI to create legal documents such as employment contracts, NDAs, lease agreements and freelance contracts. The user enters basic details (document type, parties, terms and date) and LegalEase produces a structured, editable legal document that can be downloaded as **.TXT**, **.DOCX** or **.PDF**.

2. Objectives

- Make legal document drafting simple for people without a legal background.
- Reduce the time and cost of drafting basic legal documents.
- Let users preview, edit and export the final document.
- Provide branded, professionally formatted output (logo, headings, footer).

3. Technology Stack

Component	Technology
Frontend	Streamlit
Backend	FastAPI, Uvicorn
AI Model	Google Gemini 1.5 Pro (google-generativeai)
Document Export	python-docx, FPDF, Pillow
Configuration	python-dotenv (.env file)
Language	Python 3.10+

4. System Architecture

```
User (Streamlit UI)
  | JSON request
  v
FastAPI Backend (main.py, routes.py)
  |
  v
Gemini Document Generator (ai_core/gemini_generator.py)
  |
  v
Gemini 1.5 Pro API
  |
  v
Generated text -> Preview / Edit -> Export (.txt, .docx, .pdf)
```

- **Frontend:** collects inputs, shows the preview, allows editing and downloads.
- **Backend:** receives the request, validates it and calls the AI module.
- **AI Core:** builds a structured prompt and sends it to Gemini.
- **Formatting modules:** convert the text into DOCX, PDF, TXT and HTML preview.

5. Project Structure

```
LEGALEASE/
|-- ai_core/
|   |-- gemini_generator.py
```

```
|  |-- generator.py
|-- frontend/
|  |-- app.py
|-- Image/
|  |-- Logo.png
|  |-- inverseLogo.png
|-- legalEaseAPI/
|  |-- main.py
|  |-- routes.py
|-- .env
|-- config.py
|-- requirements.txt
|-- run.sh
```

6. Key Features

- 1 **AI document generation** from user inputs using Gemini.
- 2 **Editable preview:** a dark-themed HTML preview and an edit box to change the text.
- 3 **Multi-format export:** TXT, DOCX (logo, Times New Roman, terms table, footer) and PDF (logo and footer on every page).
- 4 **Text sanitising:** removes special characters and typographic quotes for clean output.
- 5 **Secure setup:** the API key is stored in a .env file, not in the code.

7. Modules

main.py creates the FastAPI app, defines the / health-check route and includes the router.

routes.py defines the DocumentRequest model (document_type, parties, terms, dates) and the POST /generate endpoint.

gemini_generator.py builds the prompt, calls generate_content and returns the generated text.

app.py (Streamlit) provides the input form, Generate and Edit buttons, the preview and the download buttons.

Utility functions

- `sanitize_text(text)`: cleans the generated text.
- `format_docx(text, doc_type)`: creates the Word file.
- `format_pdf(text, doc_type)`: creates the branded PDF.
- `format_html_preview(text)`: creates the styled preview.

8. API Reference

POST /generate

Request body:

```
{
  "document_type": "Freelance Work Contract",
  "parties": "Jane Doe (Service Provider), TechNova Inc. (Client)",
  "terms": "Work must be delivered by May 15, 2025; Payment within 7 days of invoice",
  "dates": "April 15, 2025"
}
```

Response:

```
{ "document": "generated legal text..." }
```

9. Setup and Installation

- 1 Install Python 3.10 or later.
- 2 Create and activate a virtual environment: `python -m venv venv` then `venv\scripts\activate`
- 3 Install dependencies: `pip install -r requirements.txt`
- 4 Create a `.env` file containing `GEMINI_API_KEY=your_api_key_here`
- 5 Start the backend: `uvicorn legalEaseAPI.main:app --reload`
- 6 Start the frontend in a new terminal: `streamlit run frontend/app.py`
- 7 Open `http://localhost:8501` in the browser.

10. How to Use

- 1 Enter the **Document Type** (e.g. Freelance Work Contract).
- 2 Enter the **Parties Involved** (names and roles).
- 3 Enter the **Terms & Conditions**, separated by semicolons.
- 4 Enter the **Effective Date**.
- 5 Click **Generate Document**.
- 6 Review the preview and click **Click to Edit Document** if changes are needed.
- 7 Download as **.TXT**, **.DOCX** or **.PDF**.

11. Use Cases

- A startup founder generating an employment contract for a new hire.
- A freelancer creating an NDA for a client.
- A landlord drafting a residential lease agreement.

12. Testing

- Generated documents with different document types and inputs.
- Verified the preview, editing and all three download formats.
- Checked the logo, footer, bullet formatting and terms table in the output.

13. Limitations

- AI output may contain placeholders (e.g. [State], [Address]) that must be filled in.
- Generated documents are drafts and should be reviewed by a qualified lawyer before use.

14. Future Enhancements

- Contract analysis and plain-language summaries.
- Multilingual support.
- Integration with legal databases.
- Cloud deployment (Render, Railway, Streamlit Cloud).
- User accounts and saved document history.

15. Conclusion

LegalEase shows how Generative AI can make legal drafting faster, cheaper and easier to access. By combining FastAPI, Streamlit and Google Gemini, the project delivers an end-to-end tool that generates, previews, edits and exports professional legal documents.